Resolving User3's orthogonal absolute-product question: minimum and minimizers

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Abstract

We resolve User3's question [1, question 515549, revision 1, September 25, 2026]: for every nonsingular real square matrix of positive size, the orthogonal minimum of $\rho(|PQ|\,|Q^\top P^{-1}|)$ is attained and equals one. Thus the proposed strict uniform lower bound cannot hold for any such matrix. We give a self-contained exposition of the triangularization proof in [2], then characterize every minimizer by independent row and column permutations to triangular form. For fixed P, the set of minimizers is finite and has an explicit description using row-permuted QR factorizations. The same minimum holds over the special orthogonal group. Finally, the companion absolute-row-sum balancing question [5] follows from the Yamabe–Yujobo theorem.

1 The question and its attained minimum

For $K \geq 1$, write $O(K) = \{Q \in \mathbb{R}^{K\times K} : Q^\top Q = QQ^\top = I\}$. For a matrix A, $|A|$ denotes entrywise absolute value, and $\rho(A)$ is the maximum modulus of its complex eigenvalues. Fix $P \in GL_K(\mathbb{R})$ and set

$$M_P(Q) = |PQ|\,|Q^\top P^{-1}|, \qquad Q \in O(K).$$

User3 [1] asks whether a quantity $\beta(P) > 1$ can bound $\rho(M_P(Q))$ below for every orthogonal Q, on a suitable class of nonsingular matrices, and asks for an intrinsic characterization of the infimum. The answer to the second question is the constant one; the answer to the first is negative for every individual P.

The triangularization argument already appears in the formal source [2, `result`]. We present its complete elementary proof, using a trace bound in place of spectral-radius monotonicity. Row Gram–Schmidt is equivalently QR factorization of $P^\top$, rather than of P; see [3] for the QR construction. Section 2 gives the equality cases. Section 3 relates the objective to diagonal scaling, and Section 4 settles the companion question from the same posing author.

Theorem 1. *For every $K \geq 1$ and $P \in GL_K(\mathbb{R})$,*

$$\min_{Q\in O(K)} \rho(M_P(Q)) = \inf_{Q\in O(K)} \rho(M_P(Q)) = 1.$$

An attaining matrix Q is obtained by applying ordered Gram–Schmidt to the rows of P and placing the resulting unit vectors in its columns. The same attained minimum holds over $SO(K)$.

Proof. For arbitrary $Q \in O(K)$, put $A = PQ$ and $B = Q^\top P^{-1}$. Then $AB = I$. The triangle inequality gives, for every i, j,

$$\delta_{ij} = \left| \sum_{k=1}^{K} a_{ik} b_{kj} \right| \le \sum_{k=1}^{K} |a_{ik}|\,|b_{kj}| = (M_P(Q))_{ij}.$$

In particular, $\operatorname{tr} M_P(Q) \ge K$. If $\lambda_1, \dots, \lambda_K$ are the eigenvalues, counted with algebraic multiplicity, then

$$K \le \operatorname{tr} M_P(Q) = \left| \sum_{j=1}^{K} \lambda_j \right| \le \sum_{j=1}^{K} |\lambda_j| \le K \rho(M_P(Q)).$$

Division by $K > 0$ proves the universal bound, without a diagonalizability assumption.

Let $p_1, \dots, p_K$ be the row vectors of P, viewed in $\mathbb{R}^K$. They are linearly independent. Ordered Gram–Schmidt defines

$$v_i = p_i - \sum_{j<i} \langle p_i, q_j \rangle q_j, \qquad q_i = \frac{v_i}{\|v_i\|_2}.$$

Each v_i is nonzero by independence. The vectors q_i are an orthonormal basis and $p_i \in \operatorname{span}(q_1, \dots, q_i)$. For $Q_* = [q_1 \ \cdots \ q_K]$, the matrix $T = PQ_*$ therefore satisfies

$$t_{ij} = \langle p_i, q_j \rangle = 0 \quad (i < j), \qquad t_{ii} = \|v_i\|_2 > 0.$$

Thus T is nonsingular lower triangular, and $T^{-1} = Q_*^\top P^{-1}$ is lower triangular. For completeness, forward substitution in $Tx = e_j$ gives $x_i = 0$ for $i < j$ and $x_j = 1/t_{jj}$, establishing both assertions about the inverse.

The matrix $|T|\,|T^{-1}|$ is lower triangular. In its ith diagonal entry, the first factor can be nonzero only for $k \le i$, while the second can be nonzero only for $k \ge i$. Hence

$$(|T|\,|T^{-1}|)_{ii} = |t_{ii}|\,|(T^{-1})_{ii}| = 1.$$

Its characteristic polynomial is $(x-1)^K$, so its spectral radius is one. This proves actual attainment, and therefore the minimum and infimum assertions.

Finally, for any diagonal sign matrix D, entrywise absolute values give $M_P(QD) = M_P(Q)$. Choose D with $\det D = \det Q_*$; such a choice is possible also when $K = 1$. Then $Q_* D \in SO(K)$ and still attains one. □

Consequently there is no $\beta(P) > 1$ valid for every orthogonal Q, for any nonsingular P or any nonempty class of such matrices. This settles both requests in [1], including attainment rather than just an asymptotic approximation to the infimum.

2 Every equality case and every minimizer

The distinction between a unit spectral radius and the identity matrix is essential. The following characterization makes that distinction precise. Row and column permutations in its statement are independent.

Theorem 2. *For $A \in GL_K(\mathbb{R})$, $K \ge 1$,*

$$\rho(|A|\,|A^{-1}|) = 1 \quad \Longleftrightarrow \quad \textit{SAC is lower triangular for some permutation matrices } S, C.$$

Proof. Put $M = |A|\,|A^{-1}|$. As in the preceding proof, $M \geq I$ entrywise. Suppose $\rho(M) = 1$. Every eigenvalue λ_j has $\Re\lambda_j \leq 1$, whereas

$$K \leq \operatorname{tr} M = \sum_{j=1}^{K} \Re\lambda_j \leq K.$$

Equality forces every $\Re\lambda_j = 1$; together with $|\lambda_j| \leq 1$ this gives $\lambda_j = 1$. Thus $N = M - I \geq 0$ has characteristic polynomial x^K, and Cayley–Hamilton gives $N^K = 0$.

The directed graph with an edge $i \to j$ when $N_{ij} > 0$ has no directed cycle. A positive cycle of length r would give $(N^{mr})_{ii} > 0$ for every positive integer m, contradicting nilpotence. Reverse topological ordering therefore gives a permutation S for which $M' = SMS^\top$ is lower triangular with diagonal one. Set $A' = SA$ and $B' = A^{-1}S^\top$, so $M' = |A'|\,|B'|$. For each j, choose k_j with

$$|A'_{jk_j}|\,|B'_{k_jj}| > 0;$$

such an index exists because $M'_{jj} = 1$. These indices are distinct: if $i < j$ and $k_i = k_j$, then $|A'_{ik_i}|\,|B'_{k_jj}| > 0$ contributes to the zero entry M'_{ij}, a contradiction. Furthermore, for $i < j$, nonnegativity and $M'_{ij} = 0$ imply $A'_{ik_j} = 0$, since $B'_{k_jj} \neq 0$. The permutation matrix $C = [e_{k_1} \;\cdots\; e_{k_K}]$ consequently makes SAC lower triangular.

Conversely, if $T = SAC$ is nonsingular lower triangular, then

$$|T|\,|T^{-1}| = S|A|\,|A^{-1}|S^\top$$

is lower triangular with diagonal one, as in Theorem 1. Permutation similarity preserves the spectrum, proving the converse. □

Corollary 3. *For each row permutation S, fix the factorization $SPU_S = L_S$ with U_S orthogonal and L_S lower triangular with positive diagonal. Then*

$$\arg\min_{Q\in O(K)} \rho(M_P(Q)) = \{U_S W : S \text{ a permutation matrix, } W \text{ a signed permutation matrix}\}.$$

In particular, there are at most $2^K(K!)^2$ minimizing matrices.

Proof. A signed permutation matrix has exactly one entry ± 1 in each row and column. Its absolute value is a permutation matrix, so $M_P(QW) = M_P(Q)$. If Q minimizes, Theorem 2 gives S, C with $T = SPQC$ lower triangular. Then

$$U_S^\top QC = L_S^{-1}T$$

is both orthogonal and lower triangular. Such a matrix is diagonal with diagonal entries ± 1: its last column has only its last entry nonzero, with modulus one; orthogonality makes the other entries in the last row zero, and induction completes the assertion. Thus $Q = U_S D C^\top$ for a diagonal sign matrix D, which has the required form. Conversely, $SPU_S = L_S$ and permutation similarity give $\rho(M_P(U_S)) = 1$; signed-permutation invariance gives the same for $U_S W$. There are $K!$ possible S and $2^K K!$ signed permutations, which proves the bound; different choices can produce the same matrix. □

A rational example. Take

$$P = \frac{1}{5}\begin{pmatrix} 3 & 4 \\ -1 & 7 \end{pmatrix}, \qquad Q_* = \frac{1}{5}\begin{pmatrix} 3 & -4 \\ 4 & 3 \end{pmatrix}.$$

Then $\det P = 1$, $Q_* \in SO(2)$, and

$$PQ_* = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad M_P(Q_*) = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}.$$

This is a nonidentity, nondiagonalizable minimizer of the objective. In contrast,

$$M_P(I) = \begin{pmatrix} 1 & 24/25 \\ 14/25 & 1 \end{pmatrix}, \qquad \rho(M_P(I)) = 1 + \frac{4\sqrt{21}}{25} > 1.$$

3 Relation to diagonal scaling

There is a precise link with optimally scaled condition numbers. For fixed $A \in GL_K(\mathbb{R})$, put $s(A) = \rho(|A|\,|A^{-1}|)$ and $\kappa_\infty(B) = \|B\|_\infty \|B^{-1}\|_\infty$. Rump [4, Lemma 2.2 and Theorem 3.1] proves, without irreducibility assumptions,

$$\begin{aligned} s(A) &= \inf_{D>0 \text{ diagonal}} \| D^{-1}|A^{-1}|\,|A|D \|_\infty \\ &= \inf_{D_1, D_2 > 0 \text{ diagonal}} \kappa_\infty(D_1 A D_2). \end{aligned}$$

Here the equality of the spectral radii of XY and YX follows from $\det(I - zXY) = \det(I - zYX)$, applied to $X = |A|$, $Y = |A^{-1}|$. It does not require $|A|$ to be invertible. These are infima: for $T = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ and $D = \operatorname{diag}(d_1, d_2) > 0$,

$$\| D^{-1}|T^{-1}|\,|T|D \|_\infty = 1 + 2d_1/d_2 > 1,$$

although the infimum is one. The orthogonal minimum of Theorem 1, by contrast, is attained.

4 The companion row-balancing question

User3's question 515547 [5] asks whether, for nonsingular lower triangular L, there are $Q \in O(K)$ and $c > 0$ such that $|Q^\top L^{-1}|e = ce$, where $e = (1, \dots, 1)^\top$. The answer is affirmative, including without triangularity of L.

Corollary 4. *For every $K \geq 1$ and $L \in GL_K(\mathbb{R})$, there exist $Q \in SO(K)$ and $c > 0$ satisfying $|Q^\top L^{-1}|e = ce$.*

Proof. For $K \geq 2$, the Yamabe–Yujobo theorem [6, p. 22, Theorem] states that a continuous real-valued function on the unit sphere in $\mathbb{R}^K$ takes a common value on some orthonormal basis. Apply it to $f(u) = \|u^\top L^{-1}\|_1$. This function is continuous and strictly positive on the sphere, since L is invertible. For the resulting basis $q_1, \dots, q_K$, put $Q = [q_1 \ \cdots \ q_K]$ and $c = f(q_i) > 0$. The ith component of $|Q^\top L^{-1}|e$ is exactly $f(q_i)$. Since $f(-u) = f(u)$, a column-sign change can make $\det Q = 1$ while preserving all row sums. For $K = 1$, take $Q = 1$ and $c = |L^{-1}|$. □

This settles exact balancing and hence the source's fallback question about approximate regularization. Balancing and spectral minimization need not use the same Q. For $P = L = \operatorname{diag}(1, 2)$, every orthogonal 2×2 matrix has $|Q| = \left(\begin{smallmatrix} a & b \\ b & a \end{smallmatrix}\right)$ with $a, b \geq 0$ and $a^2 + b^2 = 1$. Equal inverse-factor row sums require $a + b/2 = b + a/2$, so $a = b = 1/\sqrt{2}$. Then

$$M_P(Q) = \begin{pmatrix} 1 & 1/2 \\ 2 & 1 \end{pmatrix}, \qquad \rho(M_P(Q)) = 2.$$

Thus no balancing matrix is a spectral minimizer in this example.

Formal correspondence. The main theorem's universal lower bound and existence of an attaining orthogonal matrix are the statement of [2, `result`], at the pinned revision listed below. Its definitions use entrywise absolute values and the complex spectral radius of the real matrix. That source proves the lower bound by spectral-radius monotonicity and constructs an attaining matrix by Gram–Schmidt triangularization. The trace bound above is an alternative elementary proof. The equality classification, special-orthogonal refinement, examples and balancing corollary have the mathematical proofs given here.

Availability and license. The accompanying source package contains this article's complete LaTeX source, metadata and a standard-library rational-arithmetic check program, with build instructions. Original exposition: CC BY 4.0, `https://creativecommons.org/licenses/by/4.0/`. The MathOverflow questions are attributed to User3 under CC BY-SA 4.0 and are paraphrased here.

References

[1] User3 (MathOverflow user 600125), *A nontrivial lower bound for the orthogonally transformed absolute product*, MathOverflow question 515549, revision 1, September 25, 2026, revision GUID `CB719581-BCE3-4573-931E-15327002F861`. `https://mathoverflow.net/questions/515549`.

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[3] N. J. Higham, *What Is a QR Factorization?*, November 10, 2020. `https://nhigham.com/2020/11/10/what-is-a-qr-factorization/`.

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[5] User3 (MathOverflow user 600125), *Can orthogonal transformations balance absolute row sums?*, MathOverflow question 515547, September 25, 2026, body in revision 1, GUID `67542193-9900-47F2-A682-20344D440FF0`; revision 2 changes title capitalization. `https://mathoverflow.net/questions/515547`.

[6] Hidehiko Yamabe and Zuiman Yujobo, *On the continuous function defined on a sphere*, Osaka Mathematical Journal **2** (1950), 19–22. doi:10.18910/4429.